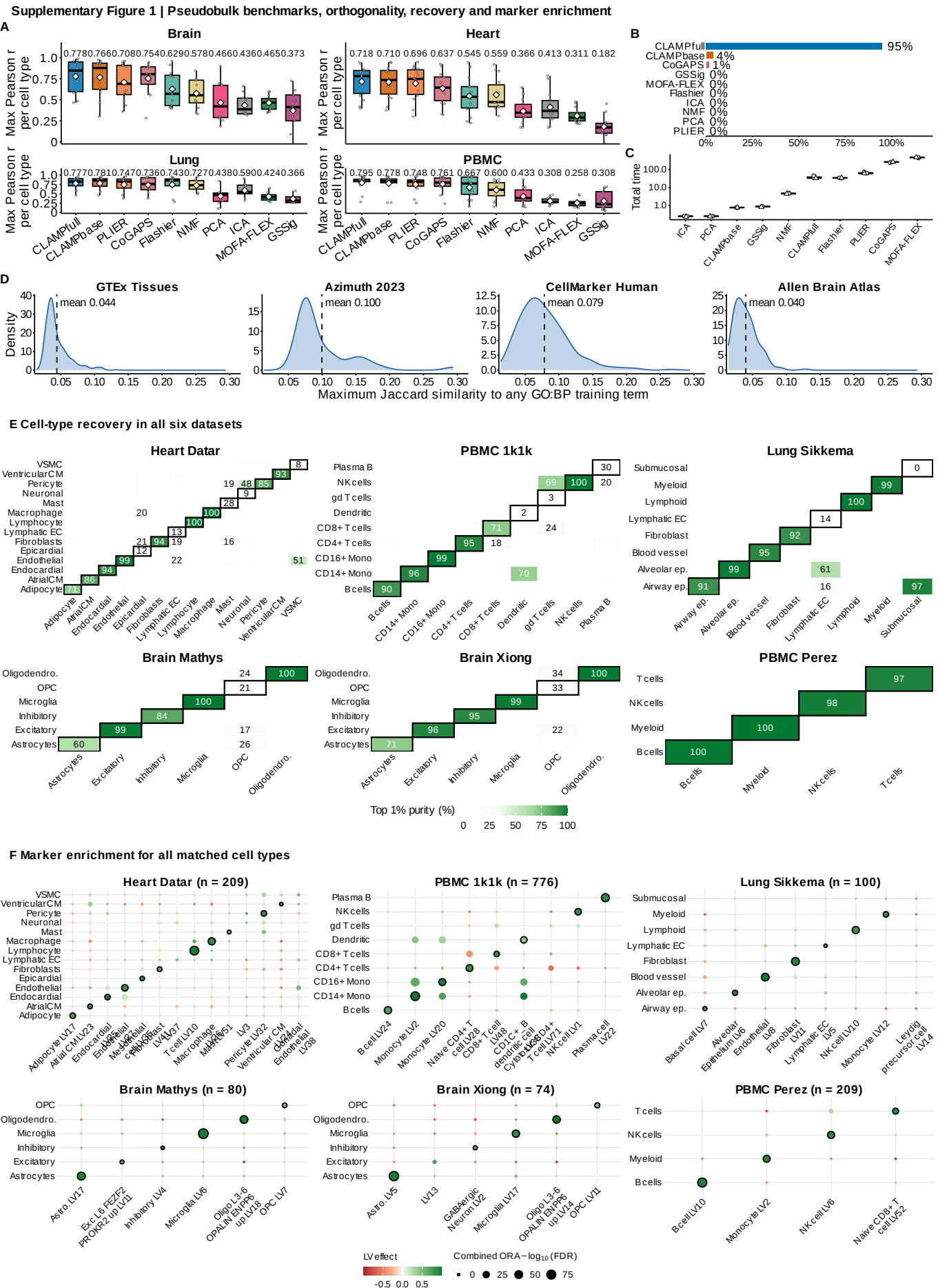

Supplementary Figure 1 | Pseudobulk benchmarks, orthogonality, recovery and marker enrichment
A
Brain
Heart
Lung
PBMC
Max Pearson r per cell type
0.778 0.766 0.708 0.754 0.629 0.578 0.466 0.436 0.465 0.373
0.718 0.710 0.696 0.637 0.545 0.559 0.366 0.413 0.311 0.182
0.777 0.781 0.747 0.736 0.743 0.727 0.438 0.590 0.424 0.366
0.795 0.778 0.748 0.761 0.667 0.600 0.433 0.308 0.258 0.308
CLAMPfull
CLAMPbase
PLIER
CoGAPS
Flashier
NMF
PCA
ICA
MOFA-FLEX
GSSig
B
CLAMPfull 95%
CLAMPbase 4%
CoGAPS 1%
GSSig 0%
MOFA-FLEX 0%
Flashier 0%
ICA 0%
NMF 0%
PCA 0%
PLIER 0%
C
Total time
ICA
PCA
CLAMPbase
GSSig
NMF
CLAMPfull
Flashier
PLIER
CoGAPS
MOFA-FLEX
D
GTEx Tissues
mean 0.044
Azimuth 2023
mean 0.100
CellMarker Human
mean 0.079
Allen Brain Atlas
mean 0.040
Density
Maximum Jaccard similarity to any GO:BP training term
E Cell-type recovery in all six datasets
Heart Datar
PBMC 1k1k
Lung Sikkema
Brain Mathys
Brain Xiong
PBMC Perez
Top 1% purity (%)
F Marker enrichment for all matched cell types
Heart Datar (n = 209)
PBMC 1k1k (n = 776)
Lung Sikkema (n = 100)
Brain Mathys (n = 80)
Brain Xiong (n = 74)
PBMC Perez (n = 209)
LV effect
Combined ORA −log10 (FDR)